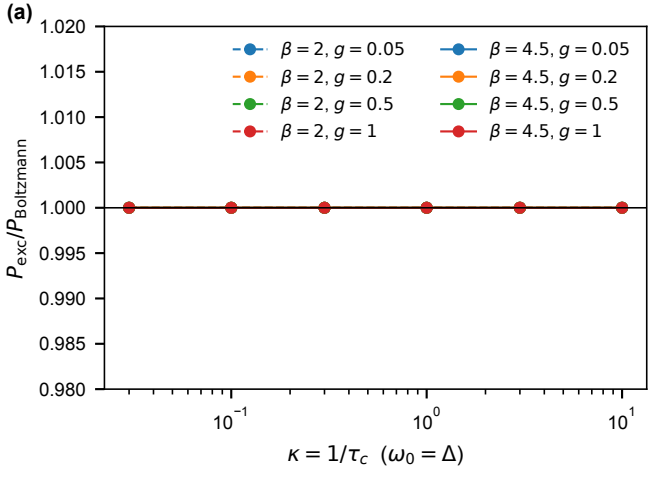
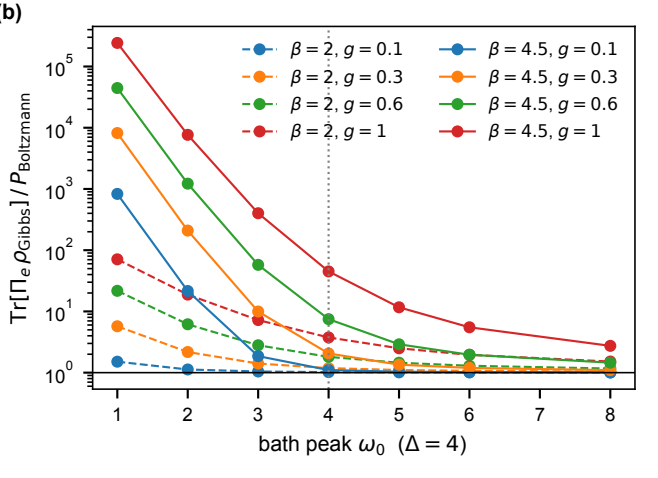

(a)
1.020
1.015
1.010
1.005
1.000
0.995
0.990
0.985
0.980
$P_{\mathrm{exc}}/P_{\mathrm{Boltzmann}}$
$\beta = 2, g = 0.05$
$\beta = 2, g = 0.2$
$\beta = 2, g = 0.5$
$\beta = 2, g = 1$
$\beta = 4.5, g = 0.05$
$\beta = 4.5, g = 0.2$
$\beta = 4.5, g = 0.5$
$\beta = 4.5, g = 1$
10^{-1}
10^{0}
10^{1}
$\kappa = 1/\tau_c \ \ (\omega_0 = \Delta)$


(b)
$\mathrm{Tr}[\Pi_e \rho_{\mathrm{Gibbs}}] / P_{\mathrm{Boltzmann}}$
10^{5}
10^{4}
10^{3}
10^{2}
10^{1}
10^{0}
$\beta = 2, g = 0.1$
$\beta = 2, g = 0.3$
$\beta = 2, g = 0.6$
$\beta = 2, g = 1$
$\beta = 4.5, g = 0.1$
$\beta = 4.5, g = 0.3$
$\beta = 4.5, g = 0.6$
$\beta = 4.5, g = 1$
1
2
3
4
5
6
7
8
bath peak ω_0 $(\Delta = 4)$